

Ten Layers from Reset Memory to Finite Cyclic Clouds

A Route Review, One Shift No-Go Theorem, and Two Surviving Physical Frontiers

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Abstract

RH-171 identified the physical packet-to-Riesz interface as the first open operator wall inside Gate A and split it into ambient realization, validated data, uniform margins, and shellwise transport. RH-172–180 investigate the ambient leaf from the actual source-memory definition rather than from an abstract equal-rank packet.

The reset memory has an exact finite history realization: $M_t = F_t^* F_t$, with a canonical polar packet and an exact rectangular cocycle $F_{t+1} = T_t F_t$. On all 130 archived snapshots these identities and the stable polar factors are numerically sound. However, none of the 120 consecutive updates passes a symmetric 0.25 primal/adjoint residual gate; the adjoint residual is always at least 0.3233. The simplest consecutive reset reducing-bundle hypothesis is therefore not supported by the finite data.

The literal square infinite-history completion is rigorously rejected as a direct determinant operator: it contains a weighted unilateral shift, has a spectral disk, and no positive power is compact or Schatten. A finite cyclic closure survives. Its reduced determinant is exactly Π_N , two copies give the RH-80 factor Π_N^2 , and a compressed rank-one edge marker separates forward from reverse orientation by exactly one. The seven RH-15 cloud rows imply a formal rank-to-cycle offset in $\{3, 4\}$, but only one row overlaps the certified reset atlas. An explicit conditional Schur theorem then lifts every doubled cycle root to a rank-two Riesz shell if a physical cycle embedding and uniform resolvent/coupling budgets are supplied.

We formalize two surviving architecture-relative frontiers. The reset- history branch is missing history-to-transfer realization, physical data, uniform margins, and shell transport. The finite-cycle branch is missing cycle calibration, cycle-to-transfer realization, and the same three physical controls. An aggregate audit records 2,600 finite formula or shell items with zero implementation/rank failures; this count is not a statistical sample size and coexists with the intentional RH-174 negative result.

Neither branch closes physical R, and the cycle branch would require a typed redesign of the reset-seeded Gate-A assembly. Q, U, Z, T and macro Gate A remain open; Gates B–E are untouched. No Hilbert–Polya or Riemann- hypothesis conclusion is claimed.

1 Coordinate before this batch

RH-161 gives a typed conditional assembly for a moving relative determinant once six obligations are met: a seed S , packet-to-Riesz lift R , cloud ledger Q , complementary Schatten limit U , canonical normalization Z , and directed marked limit T [1]. RH-162–171 then resolve the abstract proof architecture of R and arrive at

$$R = X_{\text{phys}} \wedge D_{\text{phys}} \wedge K_{\text{phys}} \wedge H_{\text{phys}}, \quad (1)$$

where X is ambient realization, D validated finite block data, K uniform Schur/graph margins, and H shellwise common-coordinate transport [2].

At that point all four leaves remained physically open. X was first because the RH-151 reset packet belongs to a source-memory Gram while the RH-80 cloud belongs to a noisy transfer operator. The present batch asks whether the special form of that Gram supplies a canonical realization and, if so, whether that realization can support a determinant.

2 Layer-by-layer result

The nine research layers before this review are summarized below.

paper	principal result	strict status
RH-172	$M_t = F_t^* F_t$ and canonical polar packet in finite history space	exact finite theorem
RH-173	$F_{t+1} = T_t F_t$, explicit $T_t^* T_t$, and orthogonal reset counterexample	exact finite theorem
RH-174	130 physical snapshots stable; zero of 120 updates pass a 0.25 two-sided residual gate	floating finite-data result
RH-175	infinite weighted history has a spectral disk and no Schatten power	rigorous branch no-go
RH-176	reduced finite cycle determinant equals Π_{L-1}	exact algebraic theorem
RH-177	double reduced cycle determinant equals RH-80's Π_N^2 factor	exact algebraic theorem
RH-178	one compressed rank-one marker separates orientation by exactly one	exact marked-trace theorem
RH-179	$r - L = \lceil h - N \rceil + 1$; seven-row gap is 3 or 4	exact arithmetic plus finite data
RH-180	explicit cyclic Schur budget preserves rank two at every root	conditional Riesz theorem

The sequence contains both positive and negative information. The negative results remove two shortcuts: Gram optimality does not imply temporal reduction, and literal infinite memory does not produce a direct Fredholm determinant. The finite cyclic branch is not an arbitrary replacement; its determinant matches the already-required geometric factor exactly.

3 The reset-history branch

3.1 What is now closed

For $S_t = A^t S_0$ and $\hat{S}_t = S_t / \|S_t\|_F$, define

$$F_t x = (\hat{S}_t x, \sqrt{\eta} \hat{S}_{t-1} x, \dots, \eta^{t/2} \hat{S}_0 x).$$

RH-172 proves

$$M_t = F_t^* F_t \tag{2}$$

and realizes every positive packet projection P_t by the canonical partial polar isometry

$$J_t = F_t P_t (P_t M_t P_t|_{\text{Ran } P_t})^{-1/2}. \tag{3}$$

This is invariant under source-coordinate gauge and packet-frame gauge [3].

RH-173 proves the exact time update

$$F_{t+1} = T_t F_t, \quad T_t(y_0, \dots, y_t) = (r_t A y_0, \sqrt{\eta} y_0, \dots, \sqrt{\eta} y_t), \tag{4}$$

with

$$T_t^* T_t = \text{operator named } \text{diag}(r_t^2 A^* A + \eta I, \eta I, \dots, \eta I).$$

Hence the finite source-memory-to-history leaf

$$X_{\text{mem} \rightarrow \text{hist}}$$

is closed exactly. This is a genuine improvement over the RH-171 status, where no canonical ambient realization was known.

3.2 What the physical audit says

The RH-174 reconstruction covers five scales, two channels, 130 packets, and 120 updates [4]. The maximum Gram and cocycle residuals are below 10^{-15} , and all stable SVD polar packets have isometry defect below 3.6×10^{-15} . The direct inverse-square-root formula becomes inaccurate at selected condition numbers up to 3.61×10^{12} , but the stable polar subspace remains well defined.

Temporal coherence is asymmetric. The primal median residual is 0.0288, but the adjoint median is 0.7130 and the adjoint minimum is 0.3233. The transported reset distance reaches 0.99999965, with 23 values at least 0.90. Therefore the data do not support treating independently reset top packets as a two-sided reducing bundle for the one-step history cocycle.

This is not an all-level no-go theorem. It leaves open lagged packets, block-time packets, separate primal/dual packets, and direct transfer-space constructions.

3.3 Why the literal infinite completion is closed off

The most direct square completion of (4) acts on infinite history by

$$\mathcal{L}(y_0, y_1, \dots) = (B y_0, q y_0, q y_1, \dots), \quad q = \sqrt{\eta}.$$

RH-175 proves

$$\sigma(\mathcal{L}) = \sigma(B) \cup \overline{D(0, q)}, \quad \sigma_e(\mathcal{L}) = \partial D(0, q), \quad (5)$$

and no positive power is compact or Schatten [5]. Thus

literal infinite history $\not\rightarrow$ direct Fredholm determinant.

Relative scattering against the shift background remains possible, but it is a different typed object.

4 The finite-cycle branch

4.1 Exact determinant model

For the length- L cycle C_L and zero-mean restriction C_L° , RH-176 proves

$$\det(I - qC_L^\circ) = \frac{1 - q^L}{1 - q} = \Pi_{L-1}(q). \quad (6)$$

The cycle differs from the nilpotent finite shift by one rank-one wrap edge of operator norm one. It converges locally/strongly to the unilateral dynamics, not in operator norm [6].

With $L = N + 1$, RH-177 defines

$$K_N = \lambda^{-1}(C_L^\circ \oplus C_L^\circ) \quad (7)$$

and proves

$$\det(I - wK_N) = \Pi_N(w/\lambda)^2, \quad (8)$$

exactly the canonical RH-80 cloud factor [7]. The complete power-trace ledger is also explicit. This closes an algebraic model leaf, not the physical cloud coefficient ledger.

4.2 Direction is additional data

Forward and reverse cycles have identical determinants and ordinary traces. RH-178 compresses the edge marker $J = |e_0\rangle\langle e_1|$ to the zero-mean space and proves

$$\mathrm{Tr}(J^\circ C_L^\circ) = 1 - L^{-1}, \quad \mathrm{Tr}(J^\circ (C_L^\circ)^{-1}) = -L^{-1}. \quad (9)$$

The orientation gap is one and is stable under operator perturbations by a trace-duality bound [8]. Therefore the appropriate finite datum is

(cloud determinant, directed marked traces),

exactly the enhanced type anticipated by RH-161.

4.3 Clock calibration is narrow but not unique

Let $h_\sigma = \log(1/\sigma)/(2 \log \lambda)$, $r_\sigma = \lceil h_\sigma \rceil + 2$, $L_\sigma = N_\sigma + 1$, and $d_\sigma = h_\sigma - N_\sigma$. RH-179 proves the exact identity

$$r_\sigma - L_\sigma = \lceil d_\sigma \rceil + 1. \quad (10)$$

On all seven RH-15 cloud rows, the right side is three or four [9]. Only $\sigma = 0.01$ overlaps the actual RH-151 reset atlas; there $r = 7$, $N = 3$, and $L = 4$. The remaining six ranks are formal clock predictions. Therefore the cycle length is constrained but not physically selected.

4.4 Conditional Riesz shells

The doubled reduced cycle has roots $\zeta_k = \rho e^{2\pi i k/L}$ of multiplicity two and half-spacing $s_L = \rho \sin(\pi/L)$. RH-180 considers a physical block candidate

$$A = \begin{pmatrix} K_L + E & B \\ C & D \end{pmatrix}$$

and circles of radius $\delta < s_L$. If $\|E\| = \varepsilon < \delta$, the complement disks are spectrum-free with contour resolvent bound d , and

$$\frac{d \|B\| \|C\|}{\delta - \varepsilon} < 1, \quad (11)$$

then every root circle encloses a rank-two Riesz projection [10]. This is the exact finite theorem needed after a physical cycle embedding is found.

5 Two surviving architecture-relative frontiers

The old leaf X_{phys} is no longer a single opaque request. There are two candidate routes to a physical finite cloud seed.

For the reset-history route define

$$R_{\text{hist}} = M_H \wedge I_{HT} \wedge D \wedge K \wedge H, \quad (12)$$

where M_H is memory-to-history realization and I_{HT} is a history-to-transfer map. For the finite-cycle route define

$$R_{\text{cyc}} = C_A \wedge C_{\text{cal}} \wedge I_{CT} \wedge D \wedge K \wedge H, \quad (13)$$

where C_A is exact cycle algebra, C_{cal} physical cycle calibration, and I_{CT} cycle-to-transfer realization.

The currently proved leaves are

$$M_H = \text{proved}, \quad C_A = \text{proved}.$$

All other displayed leaves are open.

Proposition 5.1 (Current minimal branch frontiers). *Relative to (12)–(13), the reset-history branch has missing bundle*

$$\boxed{\{I_{HT}, D, K, H\}}, \quad (14)$$

and the finite-cycle branch has missing bundle

$$\boxed{\{C_{\text{cal}}, I_{CT}, D, K, H\}}. \quad (15)$$

Neither branch is closed. The no-go theorem for the literal infinite shift does not remove the finite reset-history route or the finite-cycle route.

Proof. Each route is a conjunction. Delete the two proved leaves and collect the remaining open leaves. The RH-175 no-go object is an attempted square completion after M_H ; it is not identical to the finite map I_{HT} and is not a leaf of the finite-cycle route. $\square$

The symbols D, K, H are shared only at the architectural level. Their actual contours, block decompositions, and transport coordinates can differ between branches. A future implementation must type them branch by branch.

6 Relation to the macro Gate-A formula

The reset-history branch fits the existing RH-161 typed assembly: an eventual reset seed is lifted through R and then combined with Q , U , Z , and T . The finite-cycle branch may bypass the reset seed entirely. If it becomes primary, the Gate-A theorem must be restated with a cyclic seed type; RH-180 alone does not authorize silently substituting a cycle for S .

Consequently the safe macro status remains

$$S \text{ conditional,} \quad R, Q, U, Z, T \text{ open,} \quad \text{Gate A open.} \quad (16)$$

The exact cycle factor advances a model for Q , not the physical Q leaf. The finite edge marker advances a model for T , not the physical common-space marked limit. The distinction between algebraic model and physical identification is maintained throughout.

7 Aggregate reproducibility audit

The RH-181 script reads the archived outputs of RH-172–180. Its count of 2,600 finite items is the sum below.

layer	recorded finite items	formula/rank failures
RH-172 polar identities	192	0
RH-173 cocycle identities	160	0
RH-174 physical snapshots	130	0 identity failures
RH-175 resolvent cases	15	0
RH-176 cyclic determinants	240	0
RH-177 determinants and traces	296	0
RH-178 marked perturbations	120	0
RH-179 joined clock rows	7	0 translation failures
RH-180 matrices and root shells	1,440	0 rank failures
total	2,600	0

The root shells are nested inside the RH-180 matrices, and the rows across papers are not independent random samples. The total is therefore a reproducibility ledger, not a statistical significance claim.

The zero failure count refers to proved formula implementations and conditional rank counts. It does not erase the RH-174 physical negative result: zero of 120 updates passes the selected two-sided residual gate. That zero is a scientific outcome, not an implementation failure.

8 Recommended next physical experiment

The finite-cycle route is now algebraically cleaner, but it must be derived from data rather than reverse-engineered from the desired determinant. The next paper should construct a target-independent finite temporal clock from the physical history cocycle. A strict protocol is:

1. choose candidate lengths $L = r-3$ and $L = r-4$ before inspecting transfer eigenvalues;
2. form consecutive stable history packet frames and their reduced cocycle matrices $B_t = V_{t+1}^* T_t V_t$;
3. build the chronological product and a polar/Procrustes wrap map from the endpoint back to the seed, using only time-domain data;
4. measure the wrap residual, primal and adjoint defects, cycle-root phase error, and the rank-one orientation mark;
5. compare both integer branches on all available scales without choosing the better branch separately at each target spectral point;
6. only if one branch survives, replace floating residuals by outward matrix balls and feed them into RH-180.

If both candidate lengths fail badly, the finite-cycle physical branch is rejected while the direct transfer-space branch remains. If one survives, the next wall is a validated cycle-to-transfer isometry and shrinking L^{-1} contour budgets.

9 Longer-term roadmap

Even a successful RH-180 physical instantiation would close only a version of R. The remaining Gate-A sequence is:

1. prove that the actual finite Riesz determinant has the geometric cloud coefficient ledger Q;
2. prove a common-space $\mathcal{S}_1$ or $\mathcal{S}_2$ complement limit U;
3. prove schedule, gauge, and zero-free normalization canonicity Z;
4. transport the directed marked words and prove their limit T;
5. assemble the canonical intrinsic relative determinant.

Only after that would the program enter the scattering/unitary completion and self-adjoint questions associated with macro Gates B and C. The $T \log T$ counting law, von Mangoldt prime-power trace formula, and completed zeta divisor identity belong to Gates C–E and have not been approached by this batch.

10 Final theorem boundary

The batch proves an exact finite history realization, its exact cocycle, a branch-specific infinite-shift no-go theorem, an exact finite cyclic cloud determinant, an exact directed marker, an integer clock translation, and a conditional rank-two cyclic Riesz theorem. The finite physical audit is reported with its negative two-sided result.

It does not prove a physical history-to-transfer or cycle-to-transfer map, unique cycle calibration, outward physical block budgets, uniform all-level Riesz shells, the actual cloud

coefficient bridge, a Schatten complement limit, determinant canonicity, directed marked convergence, macro Gate A, any self-adjoint Hilbert–Polya operator, a $T \log T$ law, a prime-power trace formula, equality with the completed zeta divisor, or the Riemann Hypothesis.

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